EXTENSION BY ZERO ALONG AN OBJECT OF A RINGED SITE IS AN EXACT FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites four external items (a categories lemma, a homology section and the extension-by-zero lemma), recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Left adjoints are right exact. Stacks Project, Categories, Lemma categories-lemma-exact-adjoint (tag 0036). Gives right exactness of j_{U!} from the adjunction.
- Sectionwise description of extension by zero and exactness of sheafification. Stacks Project, Modules on Sites, Lemma lemma-extension-by-zero (tag 09B5). Supplies the explicit description of the presheaf map used in the left-exactness argument.

Source reference. Original reference 0039: sources/stacks-chow/tags.txt, source lines 126–126. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 010M: sources/stacks-chow/tags.txt, source lines 1332–1332. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03DI: sources/stacks-chow/tags.txt, source lines 4421–4421. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03EI: sources/stacks-chow/tags.txt, source lines 4457–4457. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $U \in \mathrm{Ob}(\mathcal{C})$. The functor $j_{U!} : Mod(\mathcal{O}_U) \to Mod(\mathcal{O})$ is exact.*

Proof. Since $j_{U!}$ is a left adjoint to j_U^* we see that it is right exact (see Categories, Lemma 0039 and Homology, Section 010M). Hence it suffices to show that if $\mathcal{G}_1 \to \mathcal{G}_2$ is an injective map of $\mathcal{O}_U$-modules, then $j_{U!}\mathcal{G}_1 \to j_{U!}\mathcal{G}_2$ is injective. The map on sections of presheaves over an object V (as in Lemma 03DI) is the map

$$\bigoplus\nolimits_{\varphi \in \mathrm{Mor}_{\mathcal{C}}(V,U)} \mathcal{G}_1(V \xrightarrow{\varphi} U) \longrightarrow \bigoplus\nolimits_{\varphi \in \mathrm{Mor}_{\mathcal{C}}(V,U)} \mathcal{G}_2(V \xrightarrow{\varphi} U)$$

which is injective by assumption. Since sheafification is exact by Lemma 03EI we conclude $j_{U!}\mathcal{G}_1 \to j_{U!}\mathcal{G}_2$ is injective and we win. □

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